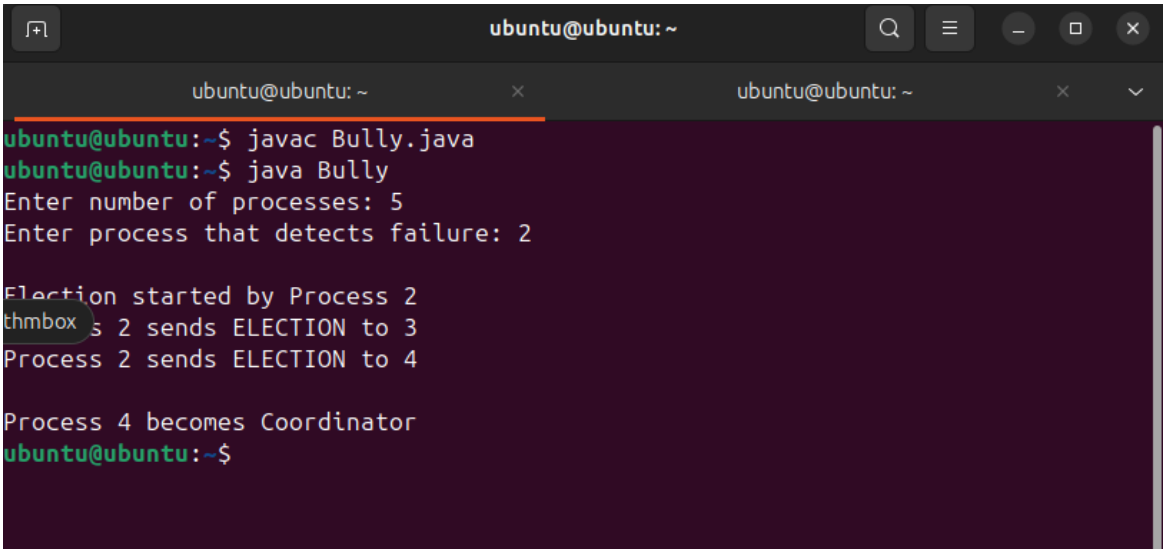
Name - Nikita Sawant
Roll No - 30
Assignment - 7

Bully.java

```
import java.util.*;

public class Bully {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      System.out.print("Enter number of processes: ");
      int n = sc.nextInt();

      boolean[] active = new boolean[n];

      // All processes active initially
      for(int i = 0; i < n; i++) {
         active[i] = true;
      }

      System.out.print("Enter process that detects failure: ");
      int initiator = sc.nextInt();

      System.out.println("\nElection started by Process " + initiator);

      int coordinator = initiator;

      // Send election message to higher processes
      for(int i = initiator + 1; i < n; i++) {
         if(active[i]) {
            System.out.println("Process " + initiator + " sends ELECTION to " + i);
            coordinator = i;
         }
      }

      System.out.println("\nProcess " + coordinator + " becomes Coordinator");

      sc.close();
   }
}
```



RingElection.java

```java
import java.util.*;

public class RingElection {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      System.out.print("Enter number of processes: ");
      int n = sc.nextInt();

      int[] process = new int[n];

      System.out.println("Enter process IDs:");
      for(int i = 0; i < n; i++) {
         process[i] = sc.nextInt();
      }

      System.out.print("Enter initiator process index: ");
      int initiator = sc.nextInt();

      ArrayList<Integer> ring = new ArrayList<>();

      int i = initiator;

      // Ring traversal
      do {
         ring.add(process[i]);
         i = (i + 1) % n;
      } while(i != initiator);

      System.out.println("\nProcesses in election: " + ring);

      // Find max ID
      int max = Collections.max(ring);

      System.out.println("Coordinator is Process with ID: " + max);

      sc.close();
   }
}
```